

Aaron Wei

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Education

University of British Columbia | MSc in Computer Science

Begins September 2026

- Supervisor: Dr. Danica J. Sutherland.

University of British Columbia | BSc in Mathematics and Computer Science

September 2021 – May 2026

- GPA: 82.8/100. Dean's list 2025, 2026. Graduate level coursework in math, computer science, and statistics.
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Work Experience

University of British Columbia | Undergraduate researcher

May 2025 – August 2026

- Conducted research in hypothesis testing with Dr. Danica J. Sutherland, resulting in a first author publication in *Transactions of Machine Learning Research* ([link](#)) and a workshop contribution at the *International Conference on Machine Learning (ICML)*.
- Generalized a foundational theorem for a class of unbiased estimators and developed these results into an effective optimization objective for kernel feature learning using deep neural networks.
- In comparison to leading methods, our optimized kernels resulted in a 17-23% increase in test power at large sample sizes when distinguishing high-dimensional image data.

Hillhouse Education | Instructor

June 2024 – November 2024

- Assisted K-12 students in learning high school subjects and basic programming in Python, in-turn increasing their conceptual understanding and exam performance.
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Projects

Undergraduate Honours Thesis

September 2024 – April 2025

- Research in kernel methods, conditional independence testing, and learning rates for function-on-scalar regression, resulting in a survey of proofs and results for kernel conditional independence testing.

Research Paper Web Scraping

July 2024

- Curated over 100 cutting-edge research papers by scraping the websites of university reading groups.
 - Accomplished the above by creating a policy compliant web scraping protocol in Python using the Scrapy library and Google's Custom Search Engine.
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Awards

- UBC BPOC Award (2026):** \$1,500 award recognizing "excellence in research and scholarly performance" for incoming master's students with a racialized identity.
 - UBC Enhanced Summer Research Experience Award (2025):** Funding for undergraduate research projects from UBC's faculty of science.
 - Advanced Machine Learning Training Network Research Award (2025):** Funding for early-career undergraduate students participating in machine learning research.
 - Livingston International Scholarship (2025):** Achievement-based scholarship for a child of a Livingston employee.
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Technical Skills

Programming languages (descending in familiarity): Python, C, MATLAB, Java, JavaScript.

Software/libraries: NumPy, Pandas, PyTorch, Scikit-Learn, OpenCV, Scrapy, Google Cloud Platform.

Extracurriculars and Volunteering

43rd International Conference on Machine Learning, Workshop on Hypothesis Testing | Reviewer

May 2026

- Provided feedback on paper submissions related to modern hypothesis testing techniques.

UBC Undergraduate Mathematics Society | Vice President of Communications

September 2025 – May 2026

- Responded to inquiries from collaborating clubs, created advertisements for club events and presented course review sessions.

UBC Uncrewed Aircraft Systems | Software team

September 2022 – December 2023

- Wrote a Python program to automate image processing tasks such as cropping, blurring and colour adjustments, and experimented with Tesseract and OpenCV filters to perform optical character detection.